

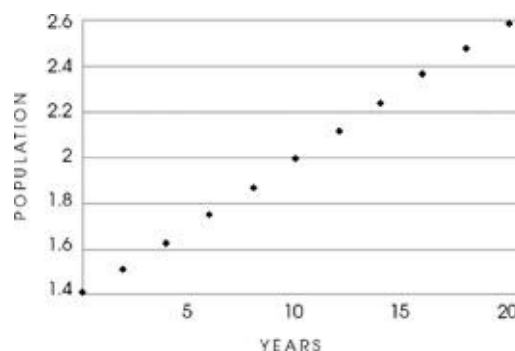
inaccurate but plain wrong as a description of reality. Their results also bother Platonified economists because they reveal that there are several ways to be irrational. Tolstoy said that happy families were all alike, while each unhappy one is unhappy in its own way. People have been shown to make errors equivalent to preferring apples to oranges, oranges to pears, and *pears to apples*, depending on how the relevant questions are presented to them. The sequence matters! Also, as we have seen with the anchoring example, subjects' estimates of the number of dentists in Manhattan are influenced by which random number they have just been presented with—the *anchor*. Given the randomness of the anchor, we will have randomness in the estimates. So if people make inconsistent choices and decisions, the central core of economic optimization fails. You can no longer produce a “general theory,” and without one you cannot predict.

You have to learn to live without a general theory, for Pluto's sake!

THE GRUENESS OF EMERALD

Recall the turkey problem. You look at the past and derive some rule about the future. Well, the problems in projecting from the past can be even worse than what we have already learned, because the same past data can confirm a theory and also its exact opposite! If you survive until tomorrow, it could mean that either a) you are more likely to be immortal or b) that you are closer to death. Both conclusions rely on the exact same data. If you are a turkey being fed for a long period of time, you can either naïvely assume that feeding *confirms your safety* or be shrewd and consider that it *confirms the danger* of being turned into supper. An acquaintance's unctuous past behavior may indicate his genuine affection for me and his concern for my welfare; it may also confirm his mercenary and calculating desire to get my business one day.

FIGURE 3



A series of a seemingly growing bacterial population (or of sales records, or of any variable observed through time—such as the